

Your Friend and Mine

The Windows Registry



What is the Registry?

- ▶ Think of as a giant 411 switchboard
- ▶ Simple idea of centralized one-stop shopping for all of Windows' needs
- ▶ Everything else is a GUI for it:
 - Windows Control Panel
 - File Associations
 - Startup Folder
- ▶ Information about WHAT and WHERE things are but not specifics on HOW to run them

Why Edit the Registry?

- ▶ Registry is the **ULTIMATE** authority
- ▶ Editing it directly allows greater control over what windows does
- ▶ Allows control over some features that don't have a GUI
- ▶ When things go bad...

Editing the Registry: The Choice is Simple

► Regedit.exe

- Designed for single user registries.
- Cleaner interface
- Available in all supported versions of Windows

► Regedt32.exe

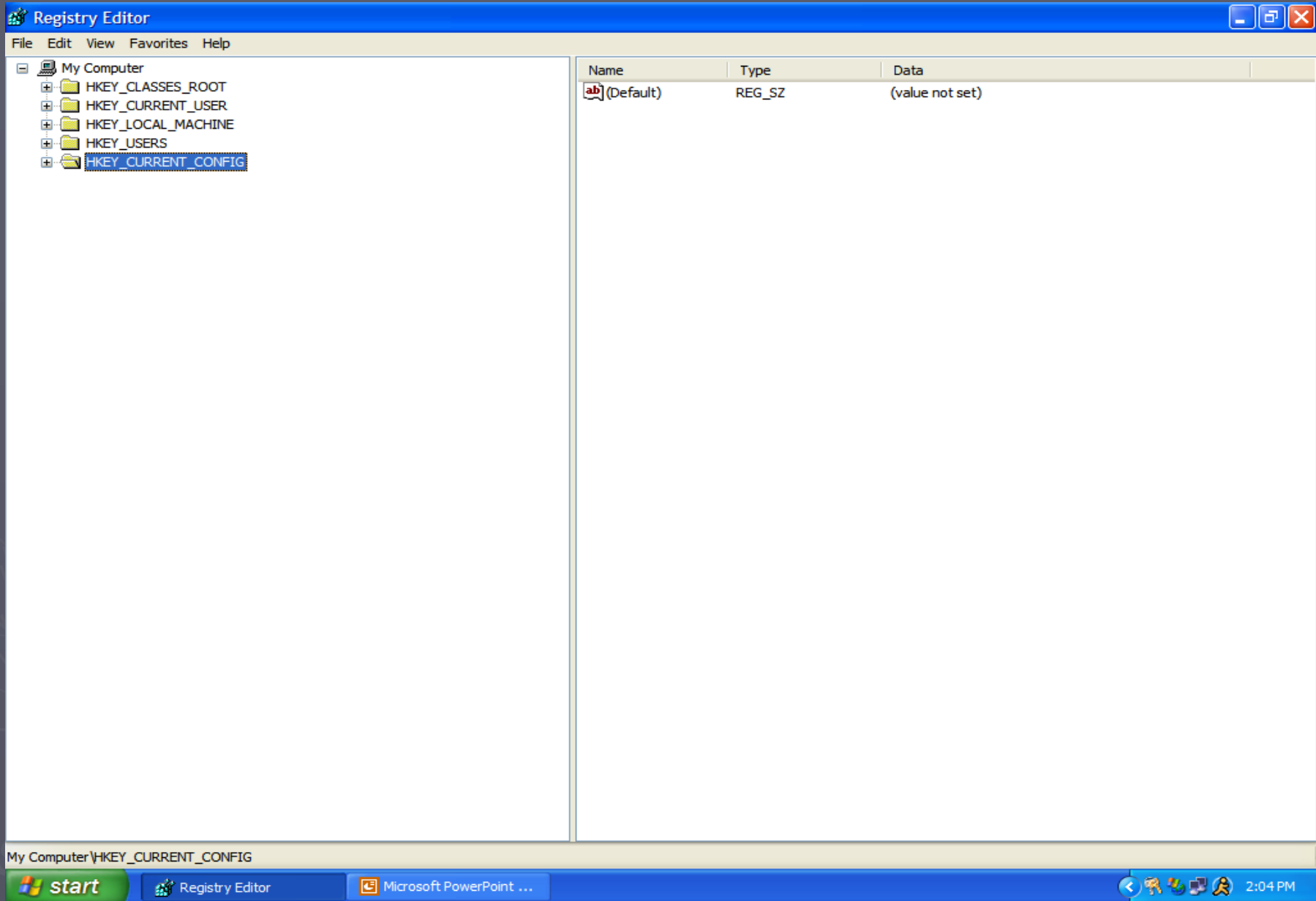
- Designed primarily for networked registries
- Available in Windows 2000, and NT
- Merged with regedit.exe in Windows XP

Registry Basics

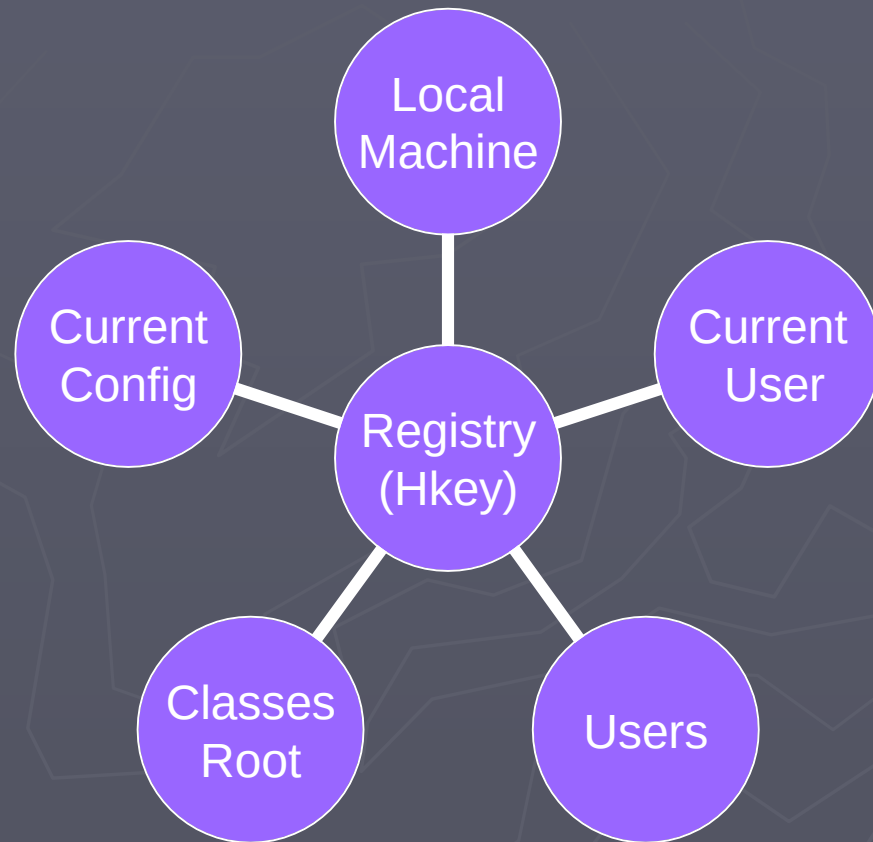
- ▶ Keys and Subkeys(Folders)
- ▶ Reg_Dword (Numbers)
 - Hexadecimal (decimal)
 - ▶ 0x00000001 (1)
 - True =1 False =0
- ▶ Reg_SZ (String)
 - Stores strings (paths to files, etc.)
 - Can be encrypted

Backup First!!

- ▶ The registry stores everything that windows knows about the computer...let that sink in.
- ▶ Backup first!
- ▶ File => Export or File => Backup
- ▶ "Scanreg /backup" and System Restore
- ▶ **MISTAKE=FORMAT!**

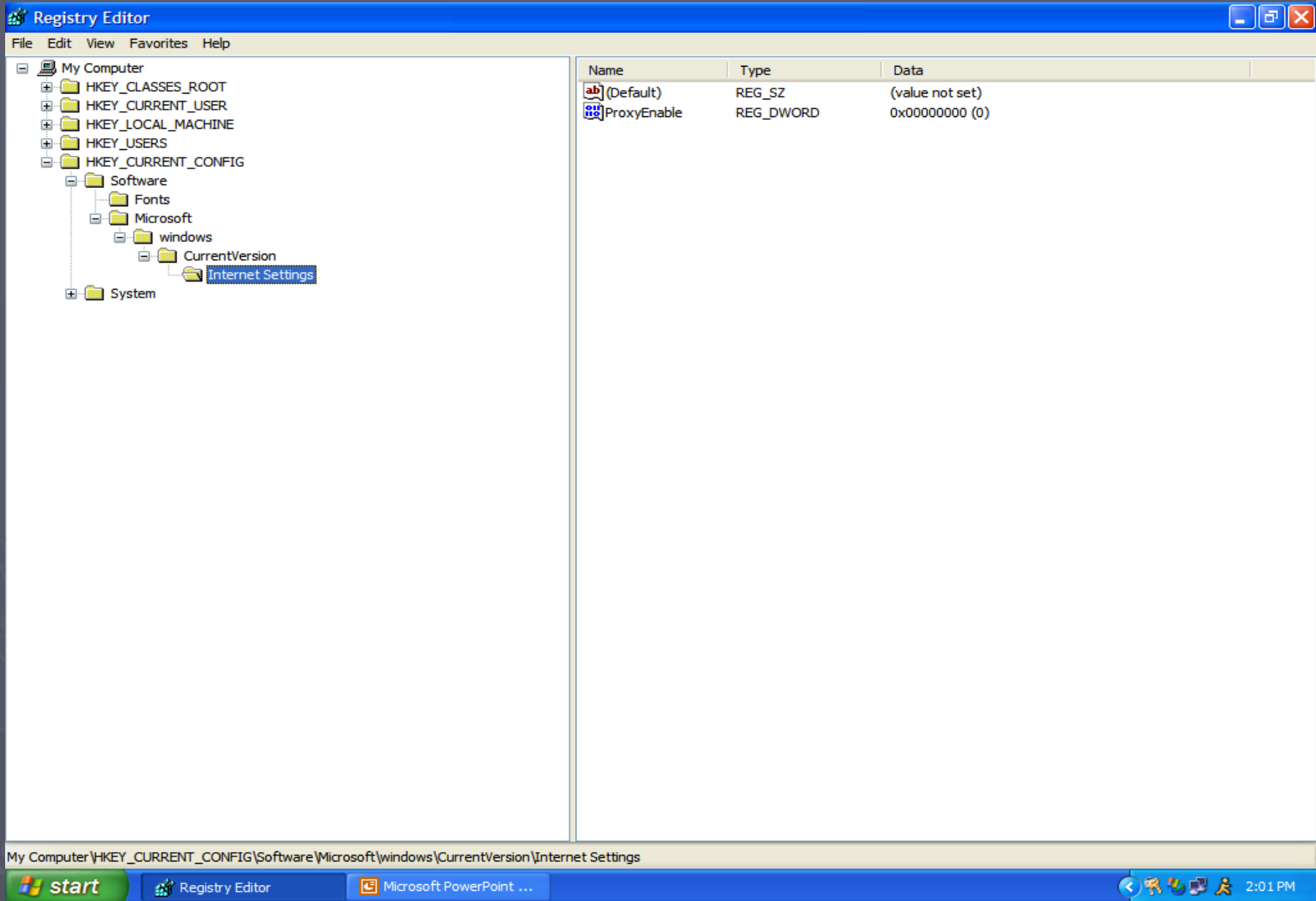


Organization of the Registry



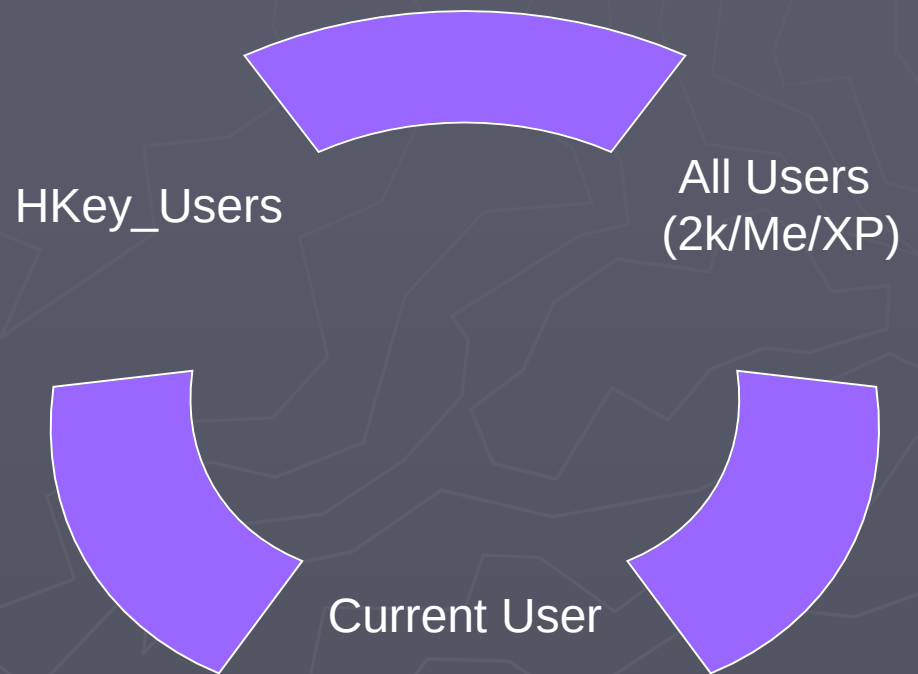
The forgotten one- *HKey_Current_Config*

- ▶ Stores temporary information about computer's settings
- ▶ Barely implemented
- ▶ *|Microsoft|Windows|CurrentVersion|InternetSettings* (proxy enable)



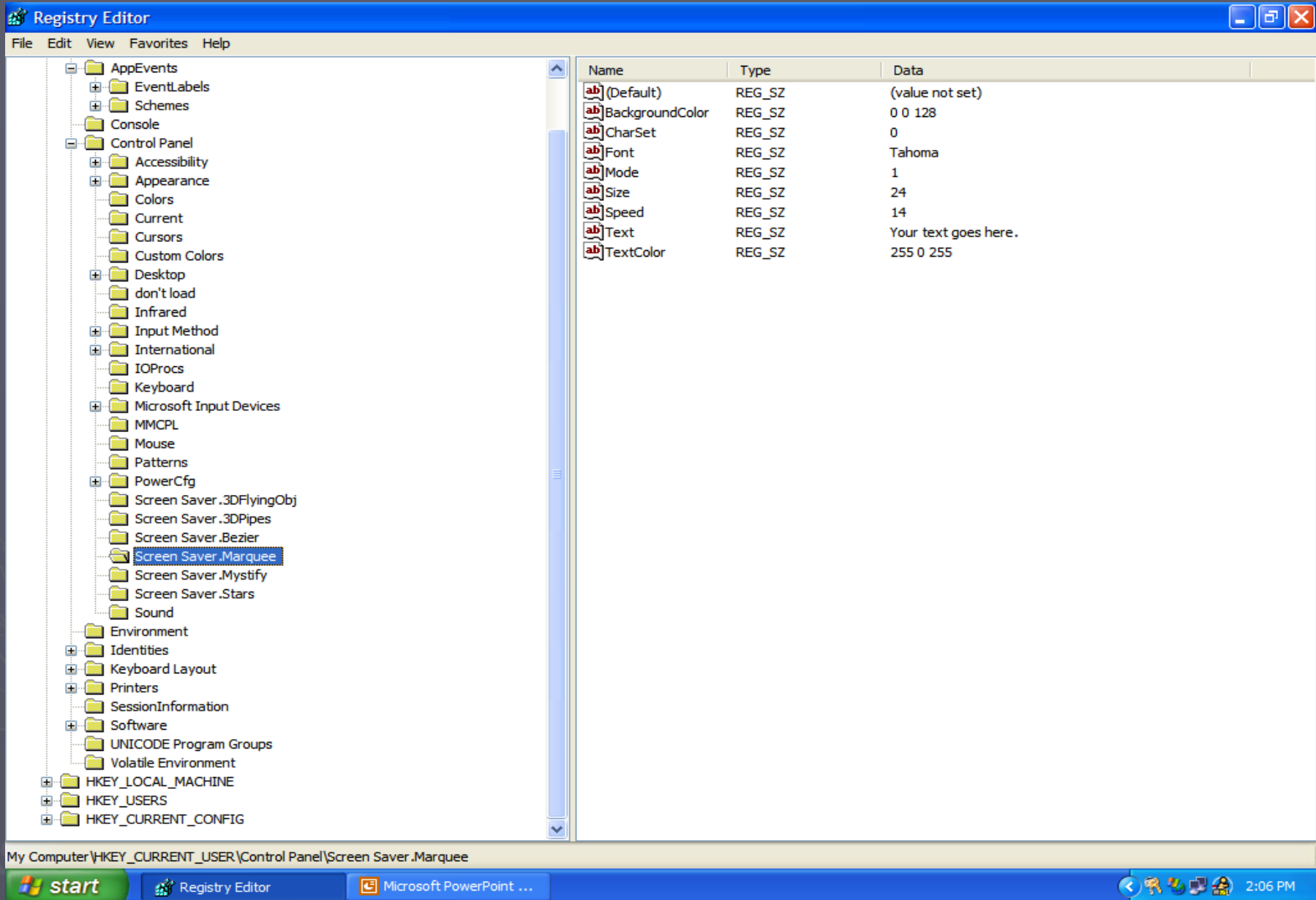
The User Database

- ▶ Personalized Settings for Windows
 - Themes
 - Accessibility
 - Preferences
- ▶ The Cycle- DB
- ▶ Saved on Exit
- ▶ Edit only Current_User



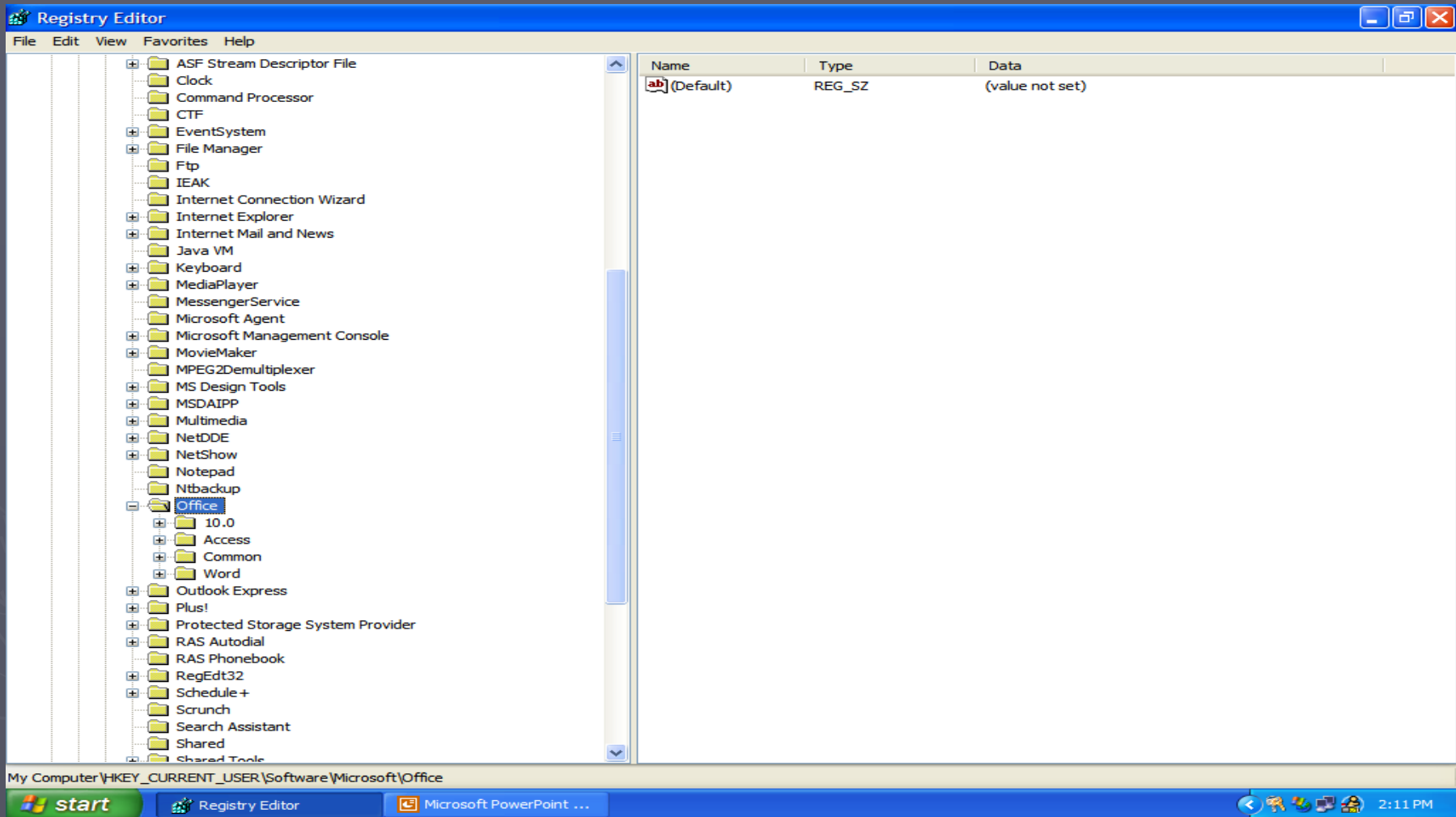
Important Stuff in HKCU

- ▶ AppEvents= Themes (Event Sounds)
- ▶ ControlPanel = duh!
 - ▶ Screen Saver
 - ▶ Desktop
- ▶ Software=User Preferences
 - ▶ \Microsoft\Office\x.y\ (office prefs)
- ▶ These keys are usually **system** safe to delete



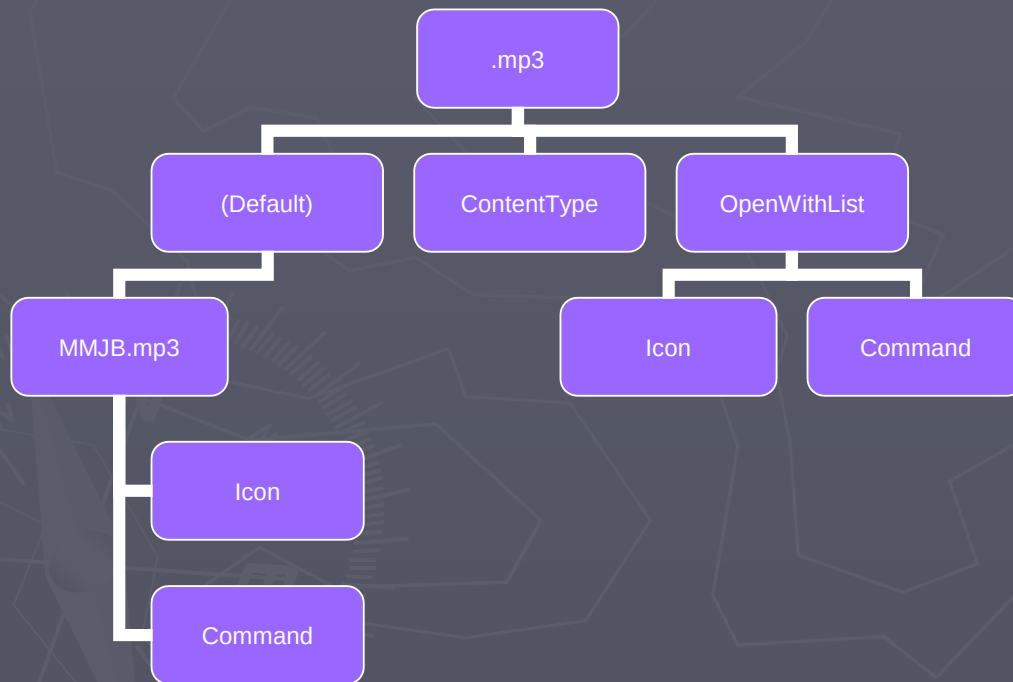
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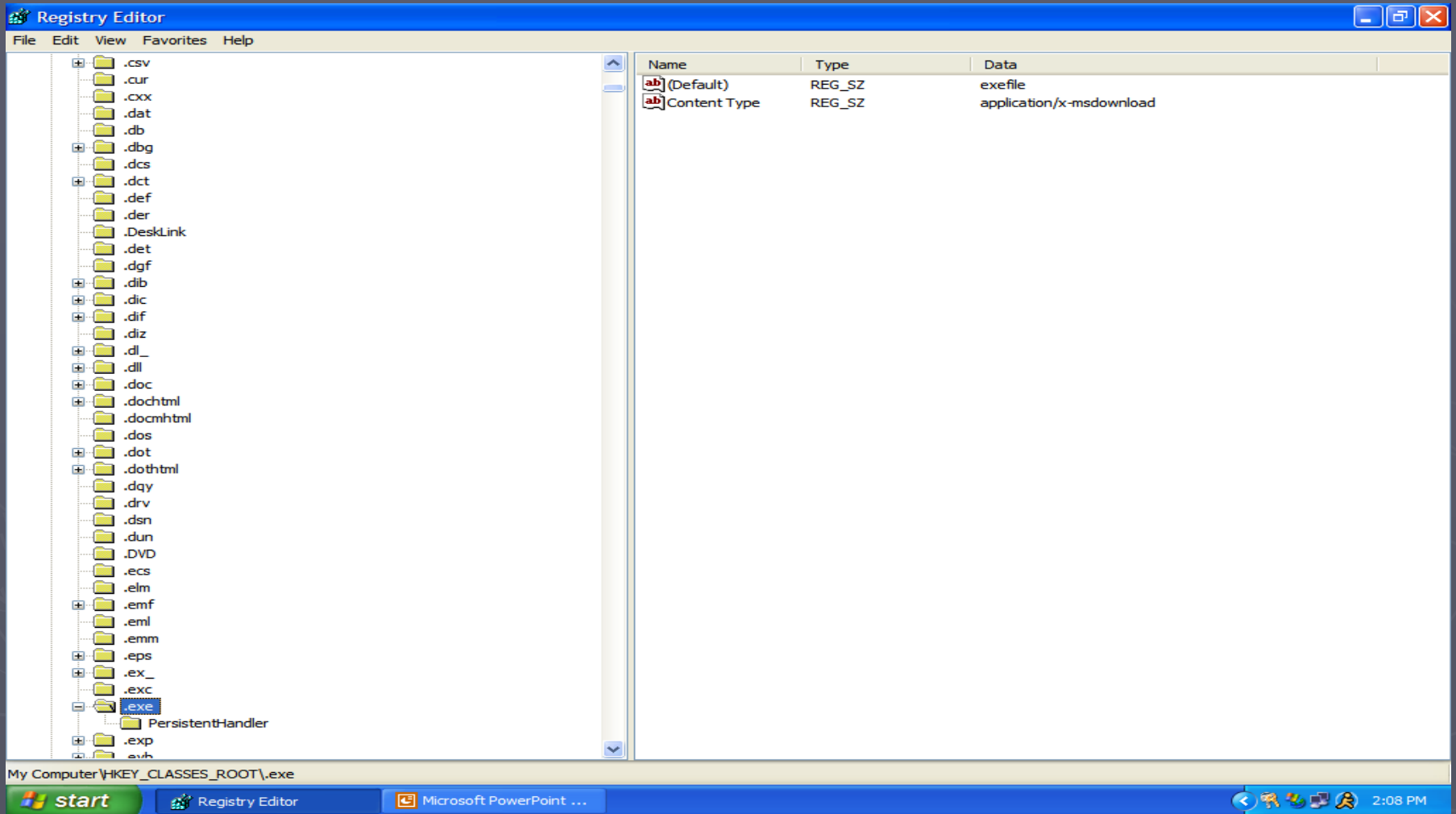


Hkey_Classes_Root:

What should I do with that?

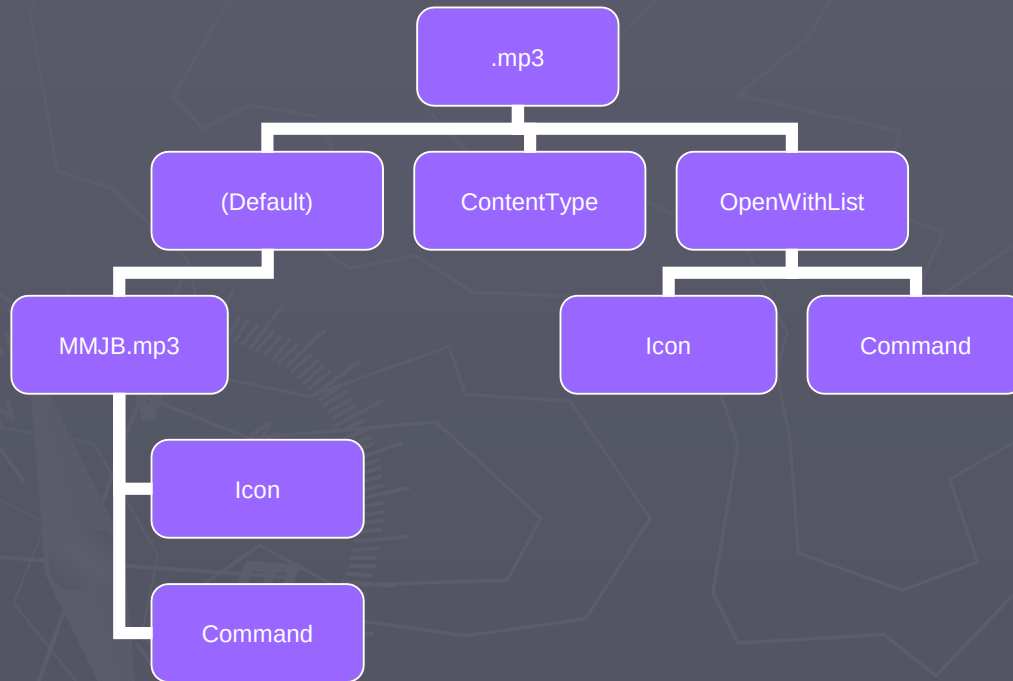


- ▶ Handles file extensions/ associations and links to methods
- ▶ Choose what opens with what (remove old apps)
 - Who wins with multiple apps
 - .mp3 => MMJB.mp3 and mp3file
- ▶ .EXE's + Viruses

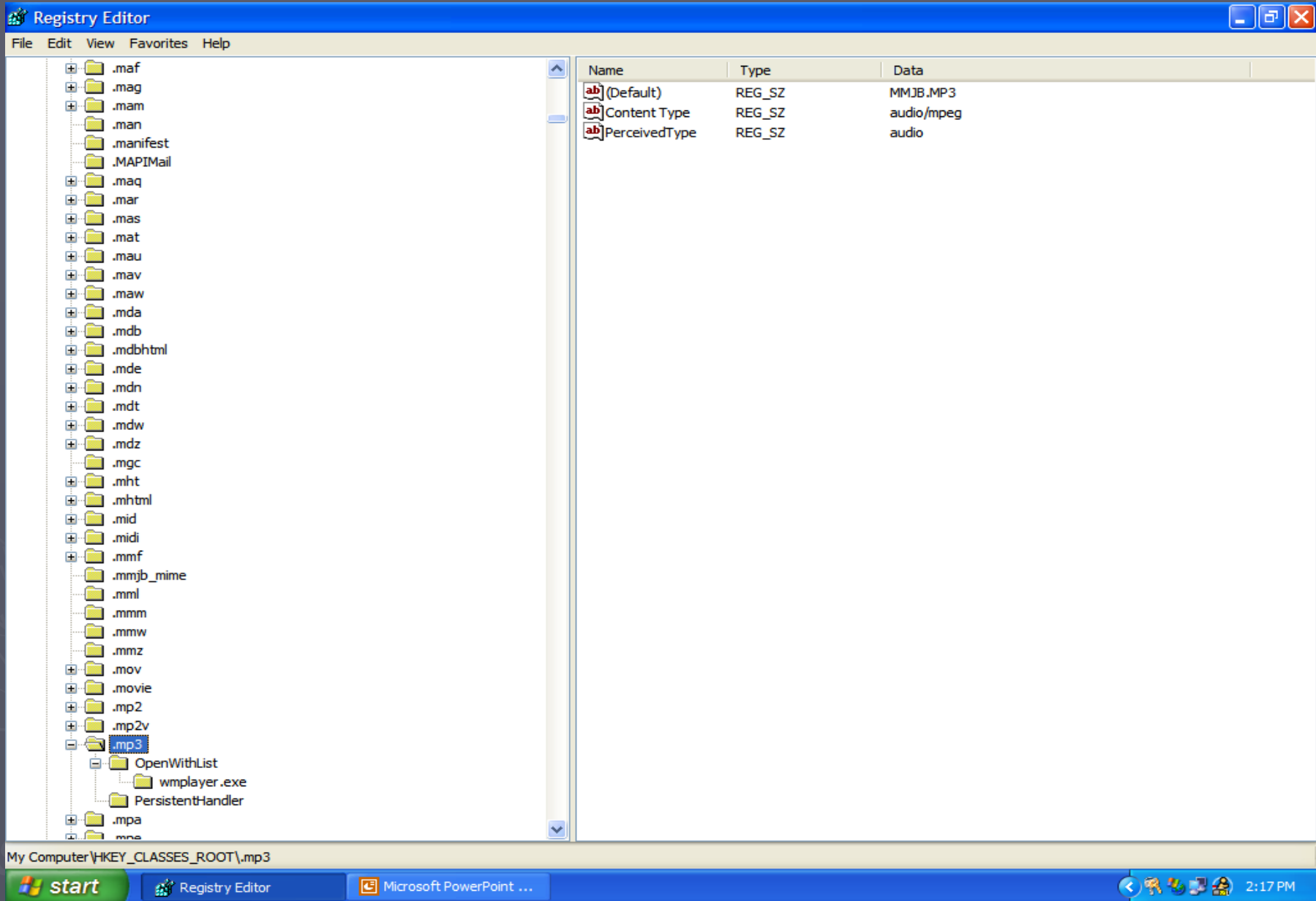


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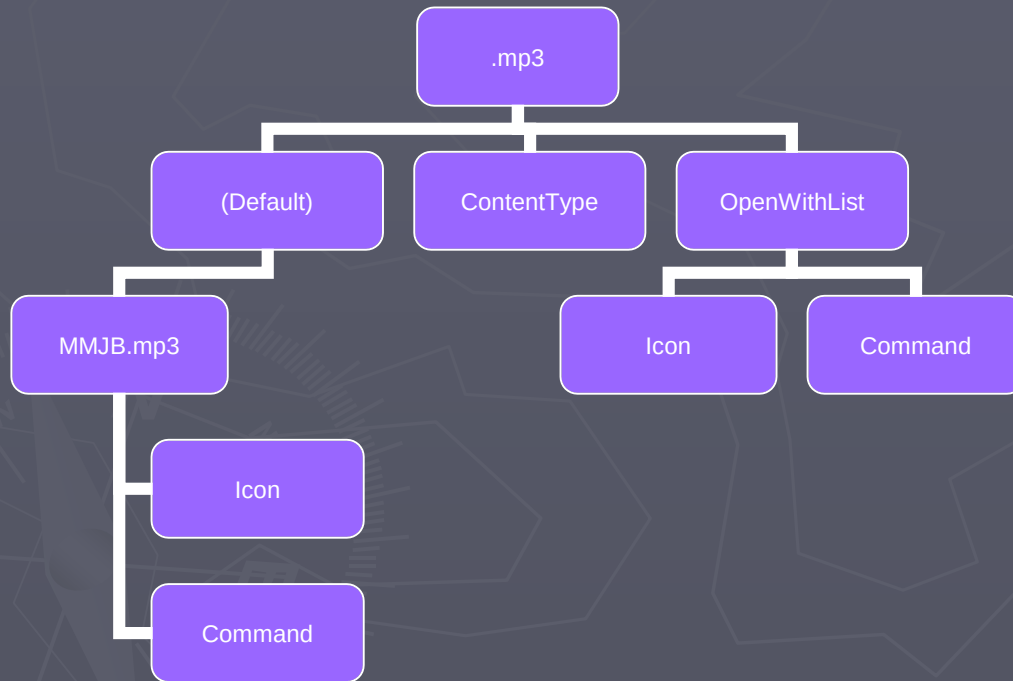


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Registry Editor

File Edit View Favorites Help

My Computer \HKEY_CLASSES_ROOT\MMJB.MP3\shell\Play

Microsoft PowerPoint - [Advanced Troubleshooting Techniques]

start Registry Editor Microsoft PowerPoint ... 2:23 PM

Registry Editor window showing the tree structure of the registry. The selected path is My Computer \HKEY_CLASSES_ROOT\MMJB.MP3\shell\Play. The right pane displays the registry value (Name, Type, Data) for the selected path.

Name	Type	Data
(Default)	REG_SZ	"C:\Program Files\MusicMatch\MusicMatch Jukebox\mmjblaunch.exe" "%1"

Registry Editor

File Edit View Favorites Help

- MMJBCTRL.LabelCtrl.1
- MMJBCTRL.PushBtnCtrl.1
- MMJBCTRL.RadBtnCtrl.1
- MMJBCTRL.TextBtnCtrl.1
- MMJBVERSION.MmjbVersionCtrl.1
- MMRadioEngine.RadioEngineObj
- MMRadioEngine.RadioEngineObj.1
- MMS
- MMST
- MMSU
- MOFL.Factoid
- MOFL.Factoid.1
- MoveBvr.MoveBvr
- MoveBvr.MoveBvr.1
- mp3file
 - DefaultIcon
 - shell
 - open
 - command
 - play
 - command
 - shellex
 - ContextMenuHandlers
 - WMPAddToPlaylist
 - WMPBurnAudioCD
 - WMPPlayAsPlaylist
- mpegfile
- MPlayer
- MS Remote
- MS Remote.1
- MS_Ink.InkObject
- MS_Ink.InkObject.1
- MSAddnDr.AddInDesigner
- MSAddnDr.AddInDesigner.1
- MSAddnDr.AddInInstance
- MSAddnDr.AddInInstance.1
- MSASR.ITN
- MSASR.ITN.1
- Msasrx.MsasrUI
- Msasrx.MsasrUI.1
- Msasrx.RecoExt
- Msasrx.RecoExt.1

Name	Type	Data
(Default)	REG_SZ	"C:\Program Files\Windows Media Player\wmplayer.exe" /Play "%L"

My Computer\HKEY_CLASSES_ROOT\mp3file\shell\play\command

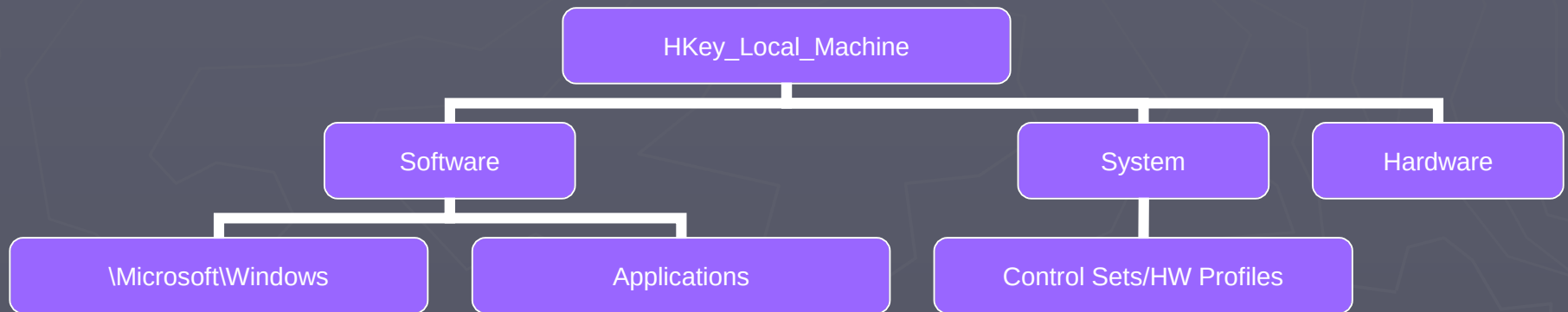


Registry Editor

Microsoft PowerPoint ...

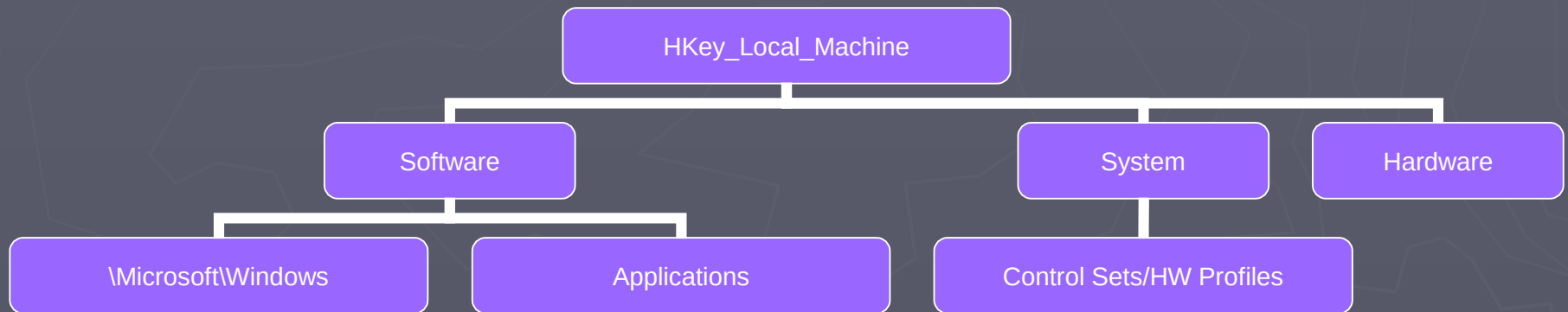
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Hkey_Local_Machine



- ▶ Software- Application Settings
- ▶ System- Control Sets
 - Control Sets = Windows HW Profiles
- ▶ Otherwise leave it alone!

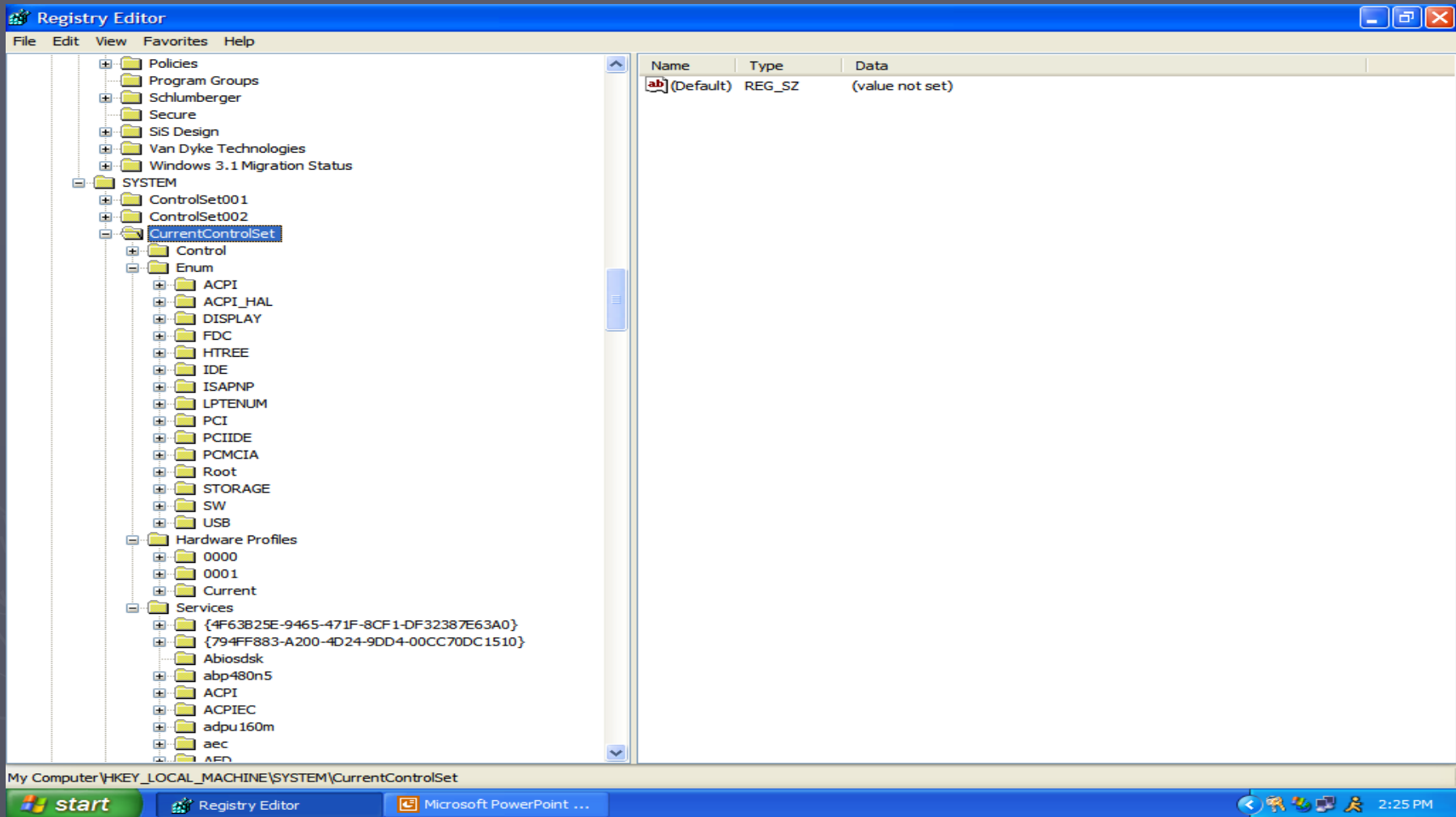
Hkey_Local_Machine



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\CurrentControlSet

- ▶ \Enum\ – same as Device Mgr
- ▶ \Control\Class- Driver Database
- ▶ HKLM\System\CurrentControlSet\Services
 - This is the source of a lot of errors
- ▶ \Services\VxD
 - Those pesky VxD's are stored here



Registry Editor

File Edit View Favorites Help

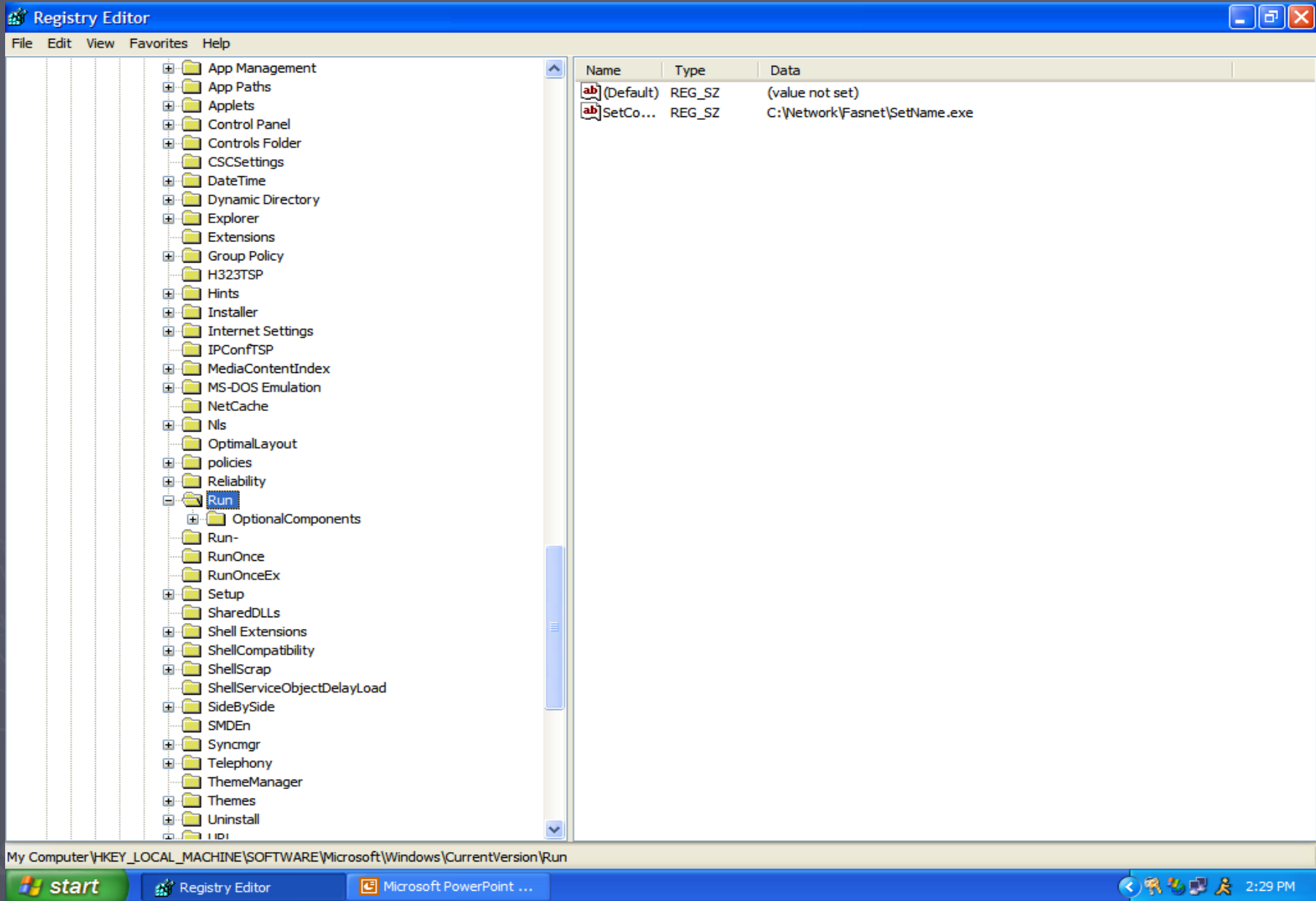
My Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip

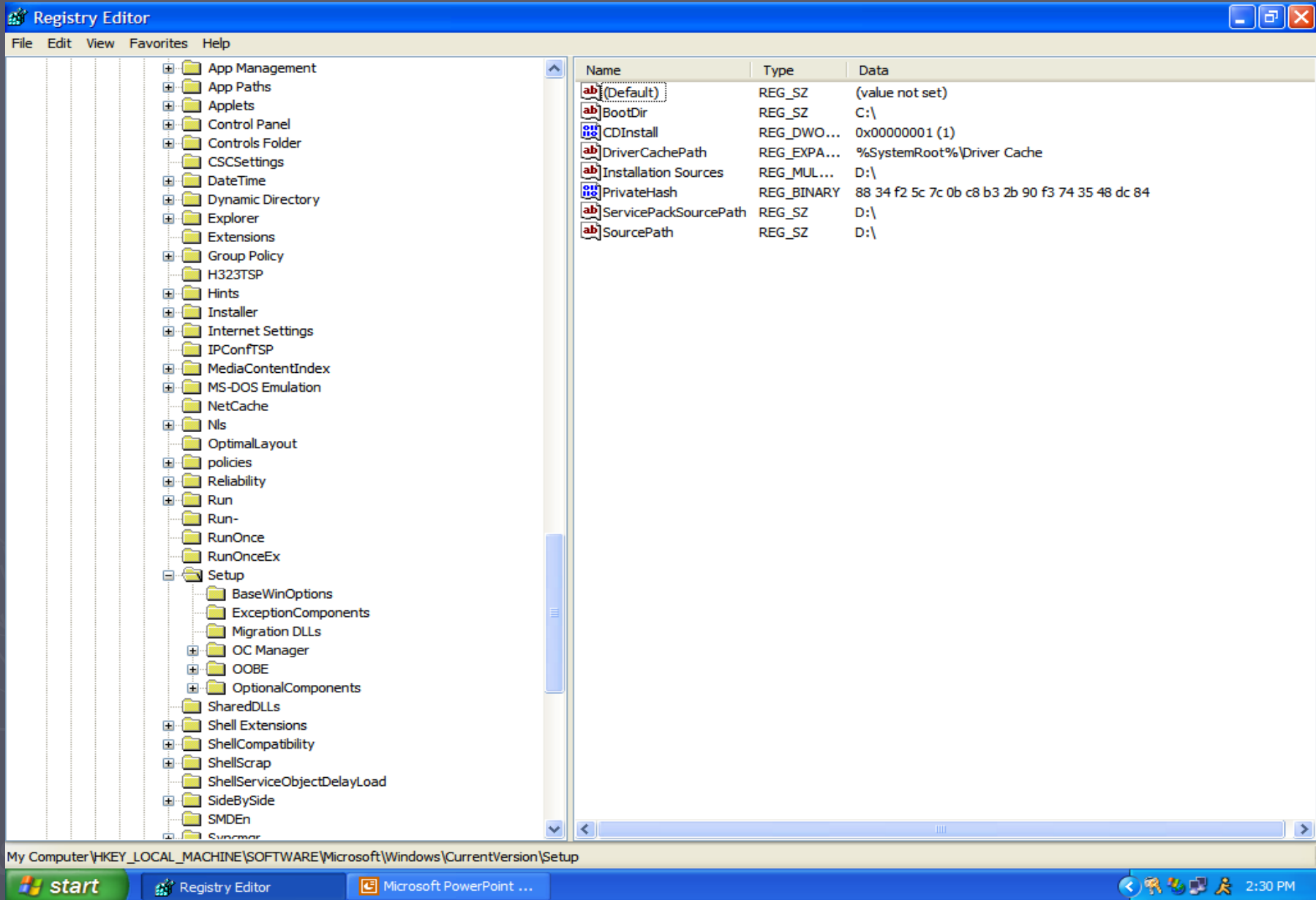
Name	Type	Data
(Default)	REG_SZ	(value not set)
Depen...	REG_MUL...	
Depen...	REG_MUL...	IPSec
Descri...	REG_SZ	TCP/IP Protocol Driver
Displa...	REG_SZ	TCP/IP Protocol Driver
ErrorC...	REG_DWO...	0x00000001 (1)
Group	REG_SZ	PNP_TDI
Image...	REG_EXPA...	System32\DRIVERS\tcpip.sys
Start	REG_DWO...	0x00000001 (1)
Tag	REG_DWO...	0x00000005 (5)
Type	REG_DWO...	0x00000001 (1)

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\Software\Microsoft\Windows\Current Version

- ▶ /AppPath – points to registered apps
- ▶ /Run/ vs /Run-/
 - Change install path
 - Finding CD keys (shhh!)
- ▶ /Setup/
 - Change install path
 - Finding CD keys (shhh!)





Registry Tricks

- ▶ Backup first!
- ▶ If you can't find it – Search!
- ▶ Copy to regedit.com if you're infected by virus.
- ▶ www.regedit.com for more info